

Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization

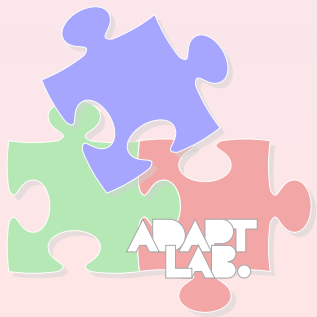
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Co-Advisor: M.Sc. Federico Bruzzone

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L-31 - Computer Science
Academic Year 2024-2025





Specialization
State of the Art and Limitations

Specialization

Problem Statement

Proposal

By Example: Implementing the Proposal

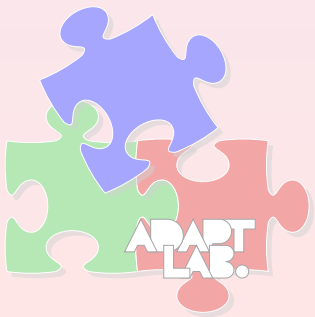
Implementation Details

Limitations

Evaluation

Conclusions

Systems programming aims to combine generic programming with type-specific optimizations.





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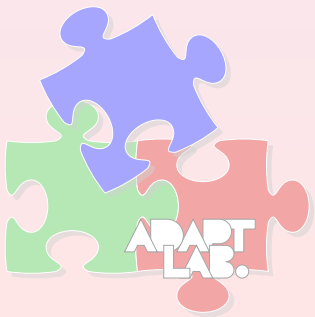
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Systems programming aims to combine generic programming with type-specific optimizations.

Mechanisms vary widely, from mature and widely used systems to experimental features, with varying degrees of success.



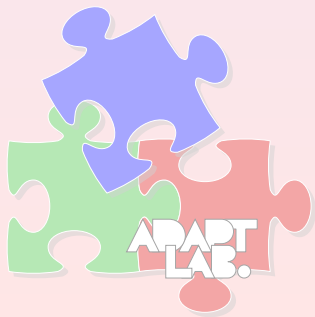


Specialization
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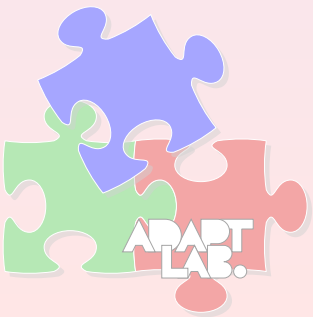
In many modern ecosystems, specialization is often unstable, forcing a tradeoff between performance and compiler stability.





Problem Statement
What's happening in Rust?

In 2015, an RFC introduced specialization to Rust as an unstable feature confined to the Nightly Channel.

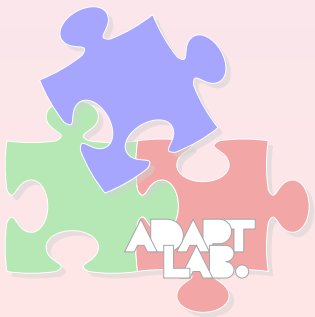




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However, due to unresolved design issues and concerns about soundness of the system, specialization is still unstable.



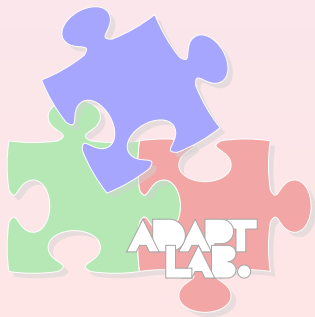


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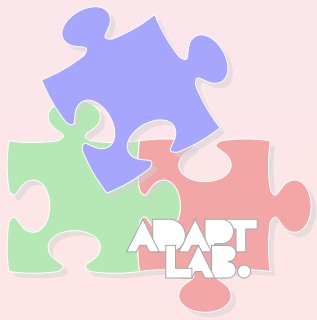
Without stable specialization, developers often resort to code duplication or complex workarounds.





Problem Statement
Motivational Example

Consider implementing a trait for different types, with a specialized implementation for a specific type.

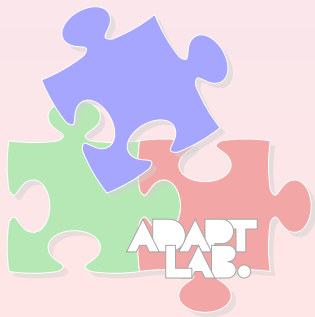




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Motivational Example

Consider implementing a trait for different types, with a specialized implementation for a specific type.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  impl MyTrait for i32 {
8      fn my_method(&self) {
9          println!("Specialized for i32");
10     }
11 }
12 fn main() {
13     10.my_method();   // should print "Specialized for i32"
14     10.5.my_method(); // should print "Generic implementation"
15 }
```



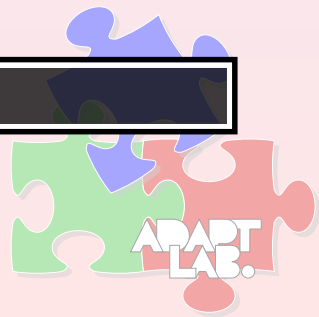


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15 }
```

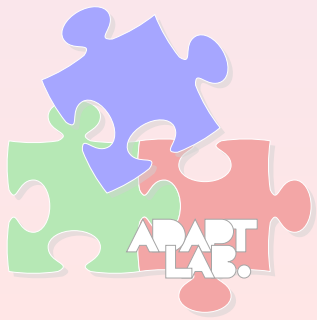
```
error[E0119]: conflicting implementations of trait MyTrait for type i32
```





Proposal
Macro-Based Specialization

Employ the powerful metaprogramming capabilities of Rust to build a macro library that simulates native specialization, with:

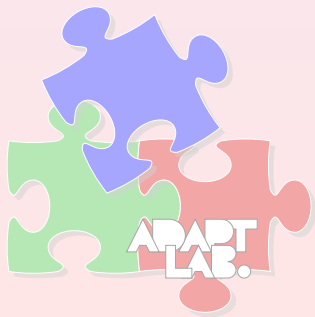




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- Native-like look and feel
- Compile-time resolution
- Sound and stable
- Zero runtime overhead
- Powerful API



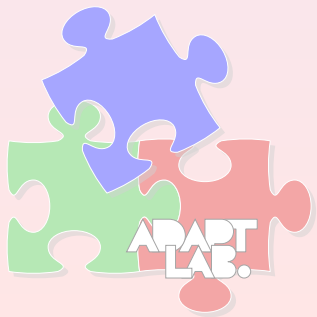


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To do this, we use two macros: `#when` and `spec!`.





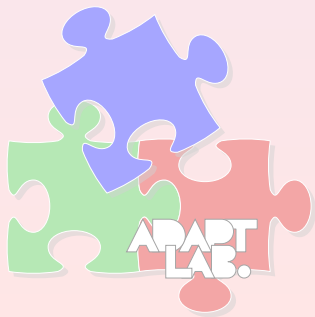
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The approach can also be applied to other languages with powerful metaprogramming capabilities that lack native specialization support.

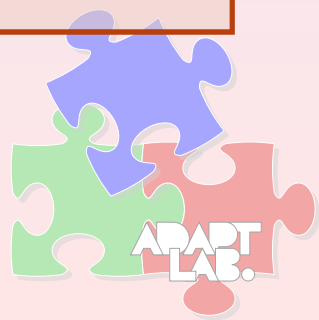




By Example: Implementing the Proposal
Motivational Example

Let's revisit the motivational example we presented earlier.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  #[when(T = i32)]
8  impl<T> MyTrait for T {
9      fn my_method(&self) {
10         println!("Specialized for i32");
11     }
12 }
13 fn main() {
14     spec! { 10.my_method(); i32 };       // prints "Specialized for i32"
15     spec! { 10.5.my_method(); f64 };     // prints "Generic implementation"
16 }
```

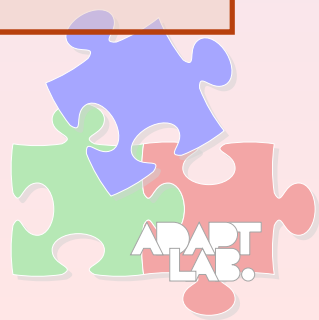




By Example: Implementing the Proposal
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Let's revisit the motivational example we presented earlier.

```
1  trait MyTrait { fn my_method(&self); }
2  impl<T> MyTrait for T {
3      fn my_method(&self) {
4          println!("Generic implementation");
5      }
6  }
7  trait MyTraitI32 { fn my_method(&self); }
8  impl MyTraitI32 for i32 {
9      fn my_method(&self) {
10         println!("Specialized for i32");
11     }
12 }
13 fn main() {
14     <i32 as MyTraitI32>::my_method(&10); // prints "Specialized for i32"
15     <f64 as MyTrait>::my_method(&10.5);  // prints "Generic implementation"
16 }
```

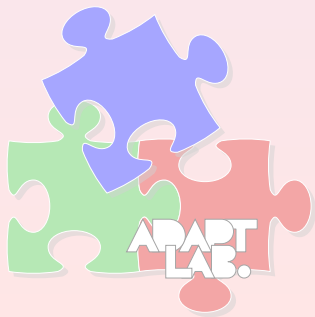




By Example: Implementing the Proposal
Supported Predicates and Forms

The following predicates are supported:

- `T = Type`
- `all(p1, p2, ...)`
- `not(p)`
- `T: Tr1 + Tr2`
- `any(p1, p2, ...)`





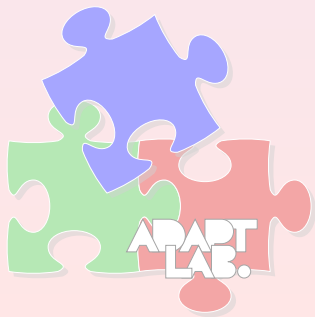
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- `T = Type`
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- `T: Tr1 + Tr2`
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We also support the following implementations forms:

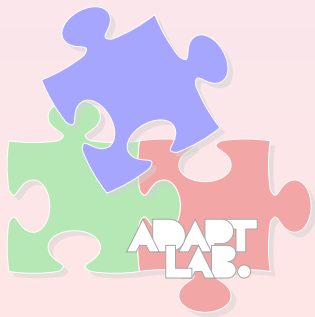
- `impl<T> Trait for T`
- `impl<T> Trait for Type<T>`
- `impl<T> Trait<T> for Type`





By Example: Implementing the Proposal
Lifetimes

Lifetimes are one of the main reasons native specialization is still unstable.

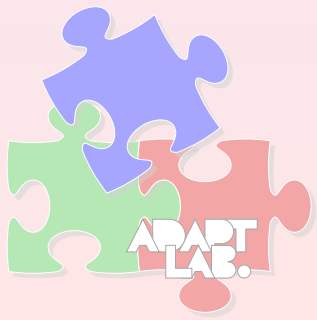




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Lifetimes bounds can be implicit or explicit.



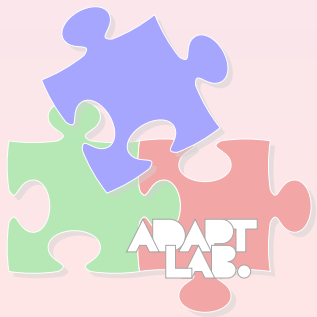


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In our belief, this is an architectural limitation of the compiler that can be overcome with proper re-design.





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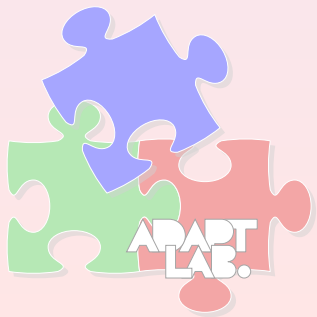
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Our proposal treats lifetimes as first-class citizens:

- `T = &'a U`
- `T: 'a`





By Example: Implementing the Proposal
Higher-Ranked Polymorphism

Specialization

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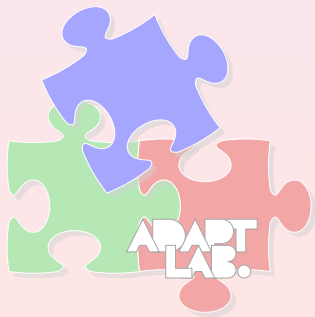
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Our macro system also supports Higher-Ranked Polymorphism.

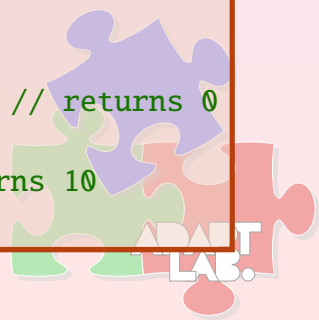




By Example: Implementing the Proposal
Higher-Ranked Polymorphism

Our macro system also supports Higher-Ranked Polymorphism.

```
1  struct S;
2  trait Trait<T, U, V> {
3      fn f(&self, t: T, u: U) -> V;
4  }
5  impl<T, U, V: Default> Trait<T, U, V> for S {
6      fn f(&self, t: T, u: U) -> V {
7          V::default()
8      }
9  }
10 #[when(all(
11     T = &str, T: 'b,
12     U = for<'a> fn(T, &'a i32) -> V
13 ))]
14 impl<'b, T, U, V> Trait<T, U, V> for S {
15     fn f(&self, t: T, u: U) -> V {
16         u(t, &7)
17     }
18 }
19 fn main() {
20     let s = S;
21     let lambda = |s: &str, n: &i32| s.len() as u32 + *n as u32;
22     let x: i32 = spec! { s.f("abc", 2); S; [&str, i32]; i32: Default }; // returns 0
23     let y: u32 = spec! { s.f("abc", lambda); S;
24       [&str, for<'a> fn(&str, &'a i32) -> u32]; u32: Default }; // returns 10
25 }
```

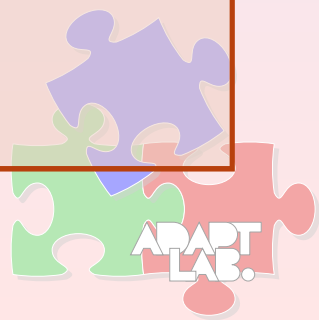




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5  impl<T, U, V: Default> Trait<T, U, V> for S {
6      fn f(&self, t: T, u: U) -> V {
7          V::default()
8      }
9  }
10 trait TraitFn<'b, V> {
11     fn f(&self, t: &'b str, u: for<'a> fn(&'b str, &'a i32) -> V) -> V;
12 }
13 impl<'b, V> TraitFn<'b, V> for S {
14     fn f(&self, t: &'b str, u: for<'a> fn(&'b str, &'a i32) -> V) -> V {
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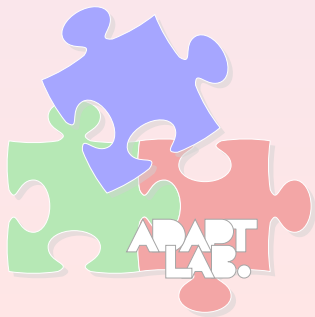




Implementation Details
Handling Conditional Implementations

Conditions defined in the `when` macro are normalized into DNF, flattening when possible, e.g.:

```
all(T = Vec<U>, any(U = i32, U = f64)) → any(T = Vec<i32>, T = Vec<f64>)
```





Meta-Monomorphic Specialization

Davide Cerato

Specialization

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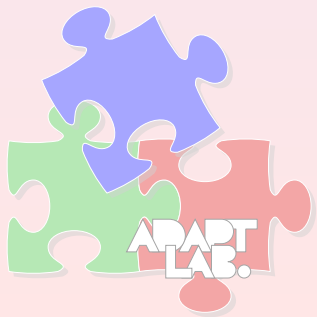
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A partial order defined over the predicates is used to determine the specificity of each implementation, e.g.:

```
T = &i32 > T = &U > T: Copy
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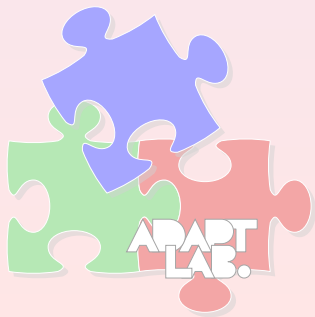
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```
T = &i32 > T = &U > T: Copy
```

We need a way to collect all the traits defined in the crate and to handle the order of their implementations. In Rust, this is done with Build Scripts, e.g.:

```
mono_spec::handle_order();
```

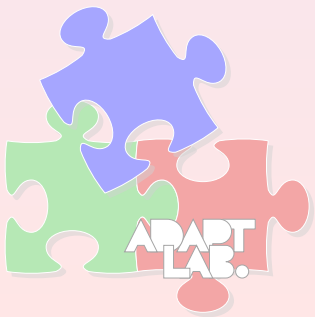




Limitations
Some Shortcomings

The usage of macros makes the tool flexible and powerful, but it also introduces some limitations and shortcomings, e.g.:

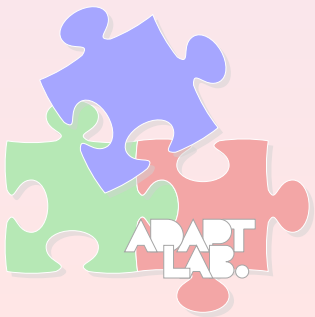
- Boilerplate
- Compile-Time Overhead
- Language Features Support
- Same Project Traits
- Warnings and Errors
- Trait Bounds





Evaluation
Approach

Once identified some patterns, we instrumented the compiler to analyze at HIR level any function definition.

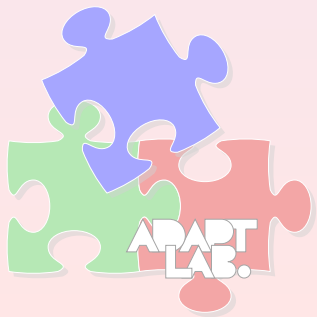




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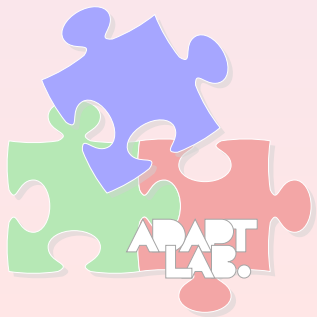
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Once identified some patterns, we instrumented the compiler to analyze at HIR level any function definition.

For each one, we created a custom AST and using heuristics, we grouped functions that could be mutually specializable.

We then calculated the tree edit distance using the ZSS algorithm and from that, we derived a similarity score:

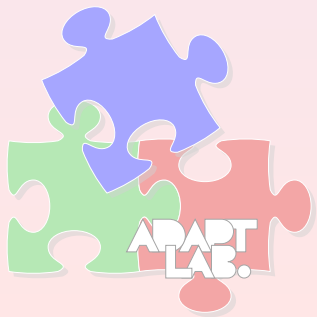
$$1 - \frac{\text{dist}}{\text{max size}}$$





Evaluation
Results

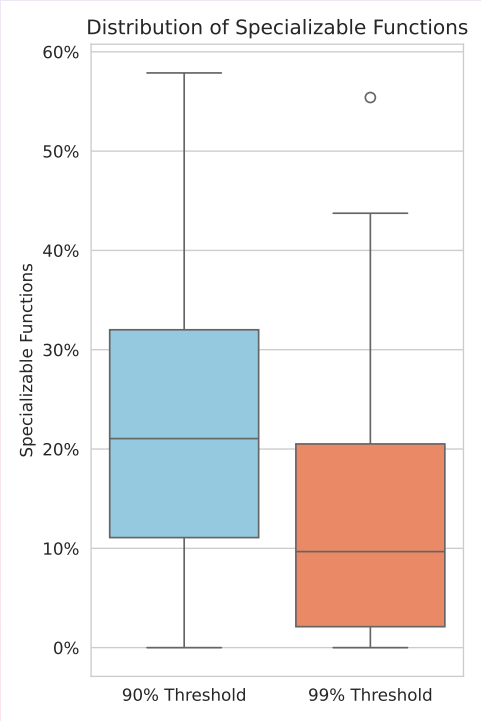
The analysis was run twice on 65 open-source Rust codebases: with a similarity threshold of 90% and 99% respectively.

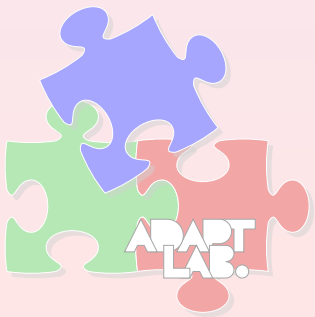




Evaluation
Results

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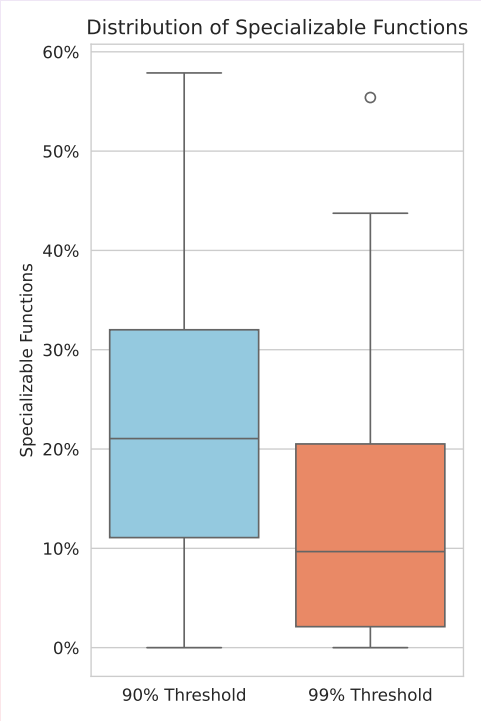


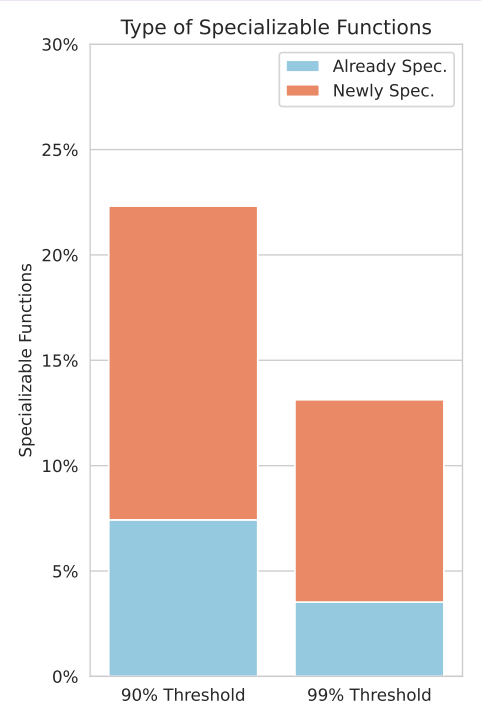


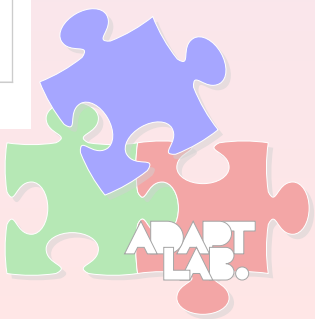


Evaluation
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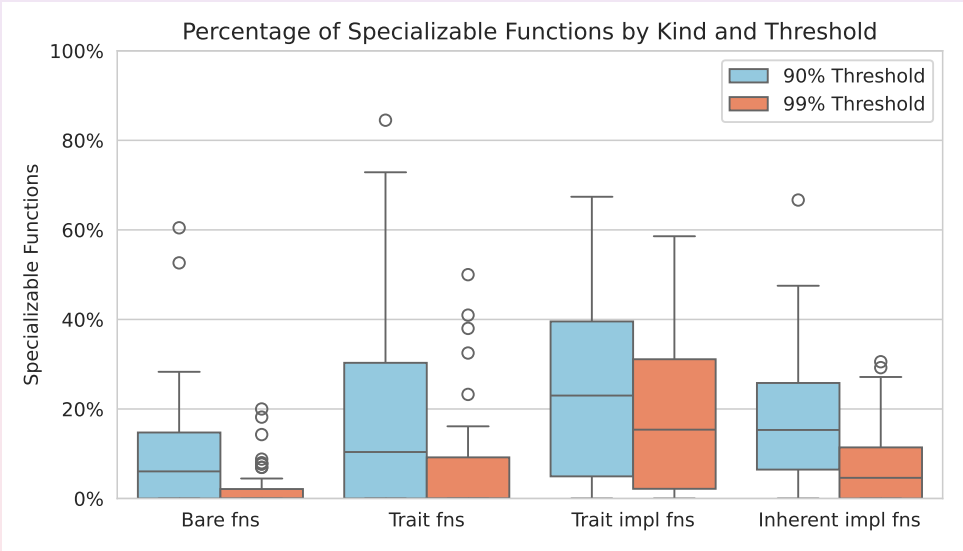


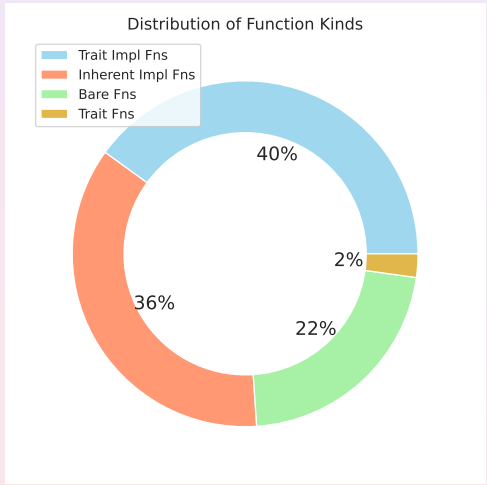


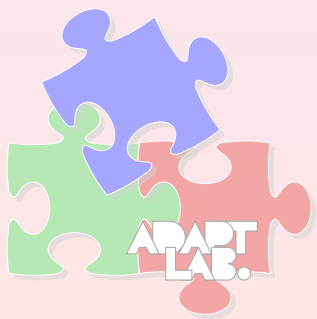


Evaluation
Results

We then analyzed the kinds of functions that were found to be specializable.









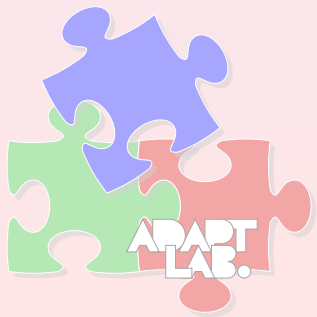
Conclusions

Interesting results:

- We are writing an article (The Soul of Meta-Monomorphizing Specializations), to be submitted to ECOOP.

Future work:

- Extend the tool to support other function kinds.





Thanks for your attention!

